## Project title

Comparing the health benefits of digital home-based resistance exercise and supervised on-site resistance exercise in cancer patients during and after treatment.

## Introduction

In Norway, ~ 35 000 individuals are diagnosed with cancer each year (www.kreftregisteret.no). Due to early diagnosis and improved treatment, three out of four patients survive the disease, and an increasing number of patients live longer with cancer (www.kreftforeningen.no, www.kreftregisteret.no). Unfortunately, many patients and survivors experience several adverse long-term side effects, which debilitate functional, mental and metabolic health (Philips & Currow 2010). Commonly reported side effects include muscle atrophy and muscular weakness, reduced physical fitness and function, fatigue, weight gain, metabolic distortions, depression and reduced quality of life (Bower 2014; Philips & Currow 2010). There is dire need for evidence-based, cost-effective, sustainable and easily accessible approaches to reduce the occurrence of these adverse health consequences during and after cancer treatment and during rehabilitation.

Resistance exercise is increasingly recognized as an important measure to counteract these treatment- and disease-related issues (Lee 2021; Lam et al 2020; Padilha et al 2016; McTiernan et al 2019; Christensen et al 2018; Keogh & MacLeod 2012). It provides relief to the range of health challenges and is accompanied by increased muscle mass and strength (Padilha et al 2016; Hasenoehrl et al 2015; Keogh et al 2012), improved functional capacity and independence (Dawson et al 2018; Keogh & MacLeod 2012) and improved markers of metabolic and lipid health (Christensen et al 2018). In addition, resistance exercise improves markers of mental well-being, such as quality of life, depression and cancer-related fatigue (McTiernan et al 2019, Hasenoehrl et al 2015; Keogh & MacLeod 2012). These benefits effectively aid cancer patients return to work earlier and lower sick leave after successful treatment, and may even improve disease-specific outcomes, including survival, disease progression, treatment tolerability and treatment responses (Christensen et al 2018).

However, resistance exercise-based therapy is only successful if the cancer patients adhere to the prescribed program. Unfortunately, exercise adherence and drop-out rates are high among cancer patients and survivors, though current estimates vary substantially, ranging between 30-93% for adherence and between 0-40% for drop-out (Bullard et al 2019; Kampshoff et al 2014). For cancer patients, exercise barriers include inconvenient location (e.g. long distance from home), exercise at an unfavourable time of the day, inadequate access to cancer-specific exercise and insufficient recommendations from healthcare providers (IJsbrandy et al 2019). Intriguingly, a recent published systematic review suggests that these issues may be possible to overcome, or at least not worsened, by using a home-based exercise approach, as a feasible, and safe alternative to on-site therapy (≥85%), despite limited face-to-face supervision and social interaction (Batalik et al 2021). According to Bullard et al (2019), home-based exercise improves flexibility and independence in the exercise planning, reduce the time used to travel, and easier implement exercise in the daily life, with higher privacy, higher protection from infections and lower costs. Home-based rehabilitation programs have also shown promising results on functional fitness, mental outcomes and treatment tolerance among cancer patients and survivors (Roberts et al 2017; Driessen et al 2016). However, previous studies have most frequently included walking, swimming or combinations of aerobic exercise and resistance exercise. Thus, there is need for more research on resistance exercise with an alternative approache in cancer rehabiliation, that potentially can target a broad range of patients, and at the same time overcome the barriers limiting participation in traditional on-site.

In Norway, «Aktiv mot kreft» has established «Pusterom» at selected hospitals (located at 21 different hospitals), which are exercise locations for cancer patients during and after treatment. Due to the Covid-19 pandemic, «Pusterommet live» was established, including home-based digital exercise sessions (www.aktivmotkreft.no). However, many cancer patients live far away from the hospitals, clinics or exercise centres offering cancer related resistance exercise, which affect their possibilities to keep up the exercise frequency. In addition, due to long-term side-effects and individual variations in the daily form after treatment, the patients need exercise flexibility. Taken together, to better understand how health-care providers and researches can optimize and deliver safe and effective resistance exercise interventions as medicine during and after treatment, there is a need for adequate studies investigating the effect of digital home-based exercise during and after cancer treatment, that in the future effectively can be outsourced independent of patients domicile.

## Hypotheses, aims and objectives

The overall research objectives is to compare the efficacy of 12 weeks digital, home-based resistance exercise program in cancer patients after treatment on functional, mental and metabolic health, compared to the same program conducted in a studio under guidance of an instructor. Selected hypotheses are presented in Table 2.
The primary aim is to compare the effect on exercise adherence between digital home-based exercise program and on-site guided program on cancer patients after treatment.
Secondary aims are to compare the two groups on several functional health outcomes, mental health outcomes and metabolic health markers (Table 1).
In addition, the project will include two subgroups of patients that will function as potential reference groups; i; a group of patients undergoing active cancer treatment and ii; eligible patients after treatment living in the Inland geographic area further than 20 km from the on-site training location. The participants in these two subgroups will be allocated to digital home-based exercise only, and not be part of the RCT. However, they will undergo the same test protocol as participants in the RCT study (Table 1). The aim is to include additional analyses concerning adherence and health benefits for patients undergoing active treatment or living too far away from on-site exercise training location.

Table 1: Overview of the included outcome domains, outcome measures, assessment methods, locations of analyses and the distributed responsibility. Abbreviations: 1RM, one repetition maximum; BBS, Bergs balance scale; DEXA, Dual-energy X-ray absorptiometry; PHQ9 questionnaire, Patient Health Questionnaire; SF-36, Short Form Health Survey; EORTC QLQ-C30, Quality of life of cancer patients; LDL, low density lipoprotein; HDL, high density lipoprotein; CRP, C-reactive protein; BP, blood pressure; INN, Inland Norway University of Applied Sciences; SI, Innlandet Hospital Trust; Nightingale Ltd, company Finland; OUS, Oslo University Hospital.

| Outcome domain | Outcome measure | Assessment method | Location of analyses | Responsibility |
|---|---|---|---|---|
| Functional health | Exercise adherence | Percentage of completed prescribed exercise sessions, | INN, SI | Solheim, Rustaden, Strøm, Sæther |
| | Maximal muscle strength | 1RM leg press (Keiser) and leg extension (Humac) | INN | Solheim, Rustaden, Hamarsland, Ma/ba students |
| | Muscular endurance | Correctly performed number of repetitions with 70% of 1RM in the exercises above and arm strength (push-up) | INN | Solheim, Rustaden, Hamarsland, Ma- and ba students |
| | Functional fitness | Six min step test, BBS | INN | Solheim, Rustaden, Hamarsland, Ma- and ba students |
| | Muscle mass | DEXA, ultrasound | INN | Solheim, Rustaden, Ellefsen, Hamarsland |
| Mental health | Fatigue | Chalder`s Fatigue questionnaire | INN | Solheim, Kolstad, Rustaden |
| | Depression | PHQ9 questionnaire | INN | Solheim, Kolstad, Rustaden |
| | Quality of life | SF-36 questionnaire, EORTC QLQ-C30 | INN | Solheim, Kolstad, Rustaden |
| Metabolic health | Visceral fat mass | DEXA | INN | Solheim, Rustaden, Hamarsland, Ellefsen |
| | Lipoprotein profile (e.g. subclasses of LDL, | Venous blood sample | Nightingale | PhD, Ellefsen, Hamarsland, |

| | | | | |
|---|---|---|---|---|
| | HDL) | | Ltd | Krook, Kolstad |
| | Markers of metabolic syndrome (e.g. fasting blood glucose, triglycerides, cholesterol, HbA1c) and inflammation (cytokines, CRP) | Venous blood sample | SI, VITAS | Solheim, Ellefsen, Hamarsland, Krook, Kolstad |
| | Other markers of the metabolic syndrome (BP, waist circumference) | BP cuff and measuring tape | SI/INN | Solheim, Ellefsen, Krook, Kolstad |

Table 2: Selected hypotheses - each representing an article included in the PhD.

| **Outcome** | **Selected hypotheses** |
|---|---|
| Functional health | Digital home-based resistance exercise program increases exercise adherence, maximal muscle strength, muscle endurance, funtional fitness and muscle mass to the same level as on-site guided program. |
| Mental health | Digital home-based resistance exercise program improve fatique, depression and quality og life to the same level as on-site guided program. |
| Metabolic health | Digital home-based resistance exercise program improves metabolic markers and reduce fat mass to the same level as on-site guided program. |

Measurements will be performed at baseline and then within two weeks after the participants have completed the program on order to compare short-term effects. Then, repeated measurments on outcomes including exercise adherence and mental health will be done at 12 months after inclusion to study long-term effects.

## Project methodology

## Project arrangements, method selection and analyses

*Study design*

The project is an assessor-blinded randomised controlled trial, where included participants living 20 km or closer to the exercise locations will be randomized to either 12 weeks of digital home-based resistance exercise (two times/week) or the same program conducted in a studio under guidance of an instructor (Fig.1).

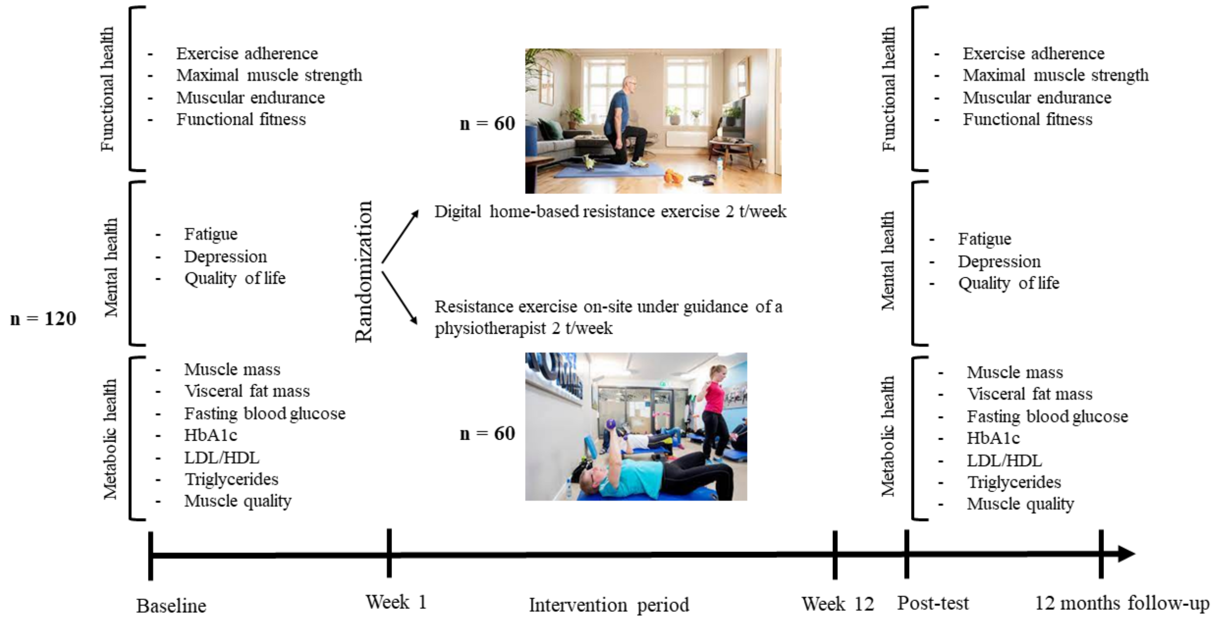


Fig.1: Study overview of the RCT.

*Participants and recruitment*

The RCT will include adult patients diagnosed with any type of cancer that have completed active hospital-based treatment (chemotherapy, radiotherapy or other infusion-based therapies) not more than five years earlier. Patients who live close to Gjøvik or Lillehammer (0-20 km traveling distance) and are willing to travel to «Pusterommet» at Gjøvik hospital or Aktivklinikken Lillehammer for exercise program will be randomized into on-site guided resistence exercise 2 t/week or digital home-based resistance exercise 2 t/week. Further, they have to have access to digital opportunities at home that makes it possible to attend the alternative treatment arm. For details about inclusion- and exclusion criteria, see Table 3. Medical doctors at the department of oncology in Innlandet Hospital Trust (SI) will recruite candidates for the study under leadership of MD/professor Arne Kolstad. Cancer patients who their doctor thinks would be able to participate in the study and who fulfill the inclusion criteria with no exclusion criteria present will be informed about the study. Moreover, social media and other media channels will be used to recruit participants. Those who show interest in partification will then be contacted by the PhD candidate to re-check inclusion- and exclusions criteria, and to give more information and receive informed consent form.

Following informed consent, study participants will be stratified into two separate groups; 1) women who have completed active treatment living approximately 0-20 km from the exercise location, 2) men living who have completes active treatment and 0-20 km from the exercise loacation.Within stratum 1 and 2 the included participants will be allocated to digital home-based and on-site exercise in a 1:1 randomized ratio (using randomization blocks of 10 participants). Randomization will be performed by a researcher not involved in the project (using R Software).
After the neccessary approvals have been gained, recruitment and the intervention will occur during a period of approximately one and a half year, to ensure sufficient number of participants. Based on how many participants we will be able to recruit, we plan two or three intervention periods. The first period will be Q3 (recruitment) and Q4 ( pilot intervention) 2023, the second and third period Q1-Q3 (recruitment) and Q4 (intervention) 2024 and 2025 respectively (Fig.2).

Table 3: Inclusion- and exclusion criteria

| **Inclusion criteria** | **Exclusion criteria** |
|---|---|
| ≥ 18 years and able to give informed consent | Co-morbidities prohibiting exercise according to protocol like serious cardic, pulmonary and other organ diseases |
| Diagnosed or previously diagnosed with any form of cancer and undergoing active treatment or completed systemic chemotherapy, immunotherapy or targeted therapies or local radiotherapy not more than 5 years earlier | Conditions caused by the cancer which prohibits exercise like previous major surgery, fractures, pain |
| Life expectancy of more than 12 months | Still undergoing active cancer treatment that is expected to confer a high risk of bone marrow suppresion that would lead to infections, bleeding and/or severly reduced physical capacity |
| Digital opportunities at home: e.g axcess to pc, and/or ipad, email adress to log on the platform for those allocated to digital training | Life expectancy of less than 12 months |
| Able to understand Norwegian language | Reduced mental capacity and not able to give informed consent and follow the program |
| Not been doing strength training regularly two times a week or more in the last three months | Travelling distance more than 20 km from home to the two exercise locations (Gjøvik and/or Lillehammer) for patients who are candidates for on-site training |

*Statistical power*
The study investigates home-based vs. on-site resistance exercise in cancer patients. The main hypothesis predicts that participants exercising home-based digital resistance exercise will achieve the same exercise adherence as participants exercising on-site. Secondary hypothesis is that home-based resistance exercise will improve several functional health outcomes, mental health outcomes and metabolic health markers to the same magnitude as on-site resistance exercise.

Power calculations were performed based on simulations of three scenarios in which an average of two sessions differences between interventions were considered a meaningful effect. The first scenario was used to assess the power to detect equivalence between interventions. Here, random samples were drawn representing adherence to a maximal of 24 sessions with an average adherence of ~ 90 and 90% in the two groups, respectively. The following scenario had average differences between groups larger than what is considered meaningful (2 sessions; see Figure 2A). Probability weights for random sampling of adherence was computed with a normal distribution centered at a specific session and standard deviation of 4, approximately corresponding to the variation reported in [@SagarraRomero2022]. A binomial model was used to model the proportion of adherence with the number of possible sessions as weights with $\alpha$ set to 0.05 to declare statistical significance. A 90% confidence interval was used for two one-tailed tests to estimate power for a test of equivalence. Power analysis showed that the study will have at least 90% power to conclude an absence of clinically meaningful difference of 5%-points (equivalence test of relative change-scores, $\alpha$ = 0.05, population SD = 7.5) if the sample size is greater than 50 participants in each group. A per group sample size of 49 participants will also have > 90% power to detect a difference between conditions corresponding to 5%-points. With an expected drop-out rate of 15-20%, the study will recruit n=60 in each group.

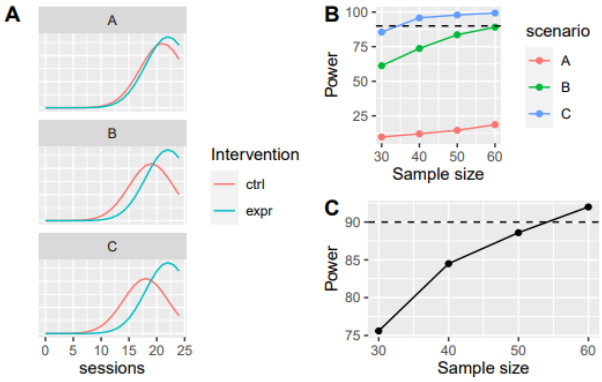


Figure 2(A): Probability weights for two groups in different scenarios. Scenario B and C corresponds to differences larger than what is considered the smallest meaningful difference. (B) shows power curves for detecting a difference given scenarios in (A) and different sample sizes. (C) shows the power to reject the null hypothesis of no difference in an equivalence test.

*Intervention*

The intervention groups will be prescribed two full-body resistance exercise sessions weekly, for a period of 12 weeks. Participants in the digital group will need an e-mail and access to a pc and/or ipad at home. They will get access to a closed group on the digital platform www.aktivmotkreft.no, for participants in the project only. This is a Norwegian cost-free digital platform, including live exercise sessions and pre-recorded exercise sessions. They will be asked to participate in pre-recorded sessions that will be available at any given time. Participants randomized to the on-site exercise group will follow the same exercise program under supervision of master- and bachelorstudents involved in the project. Participants living around the Gjøvik area will have their exercise sessions at «Pusterommet Gjøvik» at SI (with physiotherapist and aktivinstruktør Aina Strøm supervising/mentoring the students), and for participants living around Lillehammer the exercise will be held at Aktivklinkken (with physiotherapist aktivinstruktør and Lene Melgård supervising/mentoring the students). The PhD student and project group will also secure quality assurance and teaching of the students. The project group will in collaboration with «Aktiv mot kreft» develop and pre-record the exercise sessions with three different levels, ensuring modification and progression opportunities. The duration of each exercise session will be 40 minutes (see Table 4 for information about the structure). All participants will receive a resistance band, to use during some of the exercises. To document exercise adherence the instructors will register attendance at the on-site and live sessions, and the participants in the digital group will receive an exercise diary that they will be asked to fill out after each pre-recorded session. In addition, all participant will receive an short exercise diary with checkboxes to fill out after each session, with questions regarding compliance (if they followed all the exercises or parts of the session and how they experienced the session).

Table 4: Structure of the 40-minute exercise session for both intervention groups.
Abbreviations: HR, heart rate; BORG RPE intensity scale, BORG rating of perceived exerction scale.

| Minutes | What? | How? | Why? |
|---|---|---|---|
| 10 min | Warm-up | - Dynamic mobility<br>- Low impact aerobic movements to music with gradually progression to high impact in level 2 and 3. Participants exercising on-site may use ergometercycling and/or treadmill.<br>- Intensity: I-zone 1-3 (BORG RPE intensity scale 10-16, 55-87% of $HR_{max}$) | - Prepare the body for the resistance exercises (e.g. increase the internal temperature and blood flow)<br>- Prepare the mental qualities<br>- Improve physical fitness<br>- Improve cardio metabolic health factors |
| 25 min | Full-body multijoint resistance exercises using body weight, stairs, chair, bottles and resistance band | **Intensity: 8-15 repetitions x 2-4 sets, 30-60 second rest between each set**<br>- Legs: squat, lunges,step-ups, deadlift, hip thrust<br>- Chest: push-ups, chest press<br>- Back: rowing, back extension<br>- Shoulders: shoulder press, lateral raise<br>- Core: static and dynamic exercises for truncus | - Strengthen muscles and bone mineral density, increase muscle mass and lean body mass<br>- Improve activity of daily living and muscular function<br>- Decrease risk of injury<br>- Improve cardio metabolic health factors |
| 5 min | Cool-down and stretching | - Low impact aerobic movements<br>- Static stretching of large muscle groups – 30 second x 2-3 series<br>- Relaxation | - Maintain or improve mobility<br>- Achieve increased relaxation and reenergizing before returning to everyday life |

*Measurements*

All tests will be carried out in the laboratory at the Inland Norway University of Applied Sciences, Lillehammer (HINN). All test/sampling protocols are established in the project consortium, and will be conducted by a standardized manner. The PhD candidate will led and coordinate the sampling/activities, supported by bachelor and master students from HINN. The test battery (Table 1) will be performed at three time-points; before and after the intervention period, and at 12 months follow up. At each time-point, the participants will need to show up at the laboratory at HINN at one day. At test day, they will do the DEXA-scan, ultrasound, blood pressure, waist cirumference, followed by a self-composed breakfast. Thereafter, they will perform the maximal muscle strength tests, muscular endurance, six minutes step test and Bergs balance test (BBS) (in total ~ 4 hours). Questionnaires will be completed during the breakfast, and will include measurements of fatique (Chalder`s Fatique questionnaire), depression (PHQ9 questionnaire) and quality of life (SF-36 questionnaire, EORTC QLQ-C30), all of which have been validated for cancer patients, and data from the normal population are available for comparison. Information about medical history, cancer diagnosis and treatment will be registered using a questionnaire, as well as using their medical hospital journal (DIPS?)". Every fourth week during the intervention period, the participants will be asked to retrospectively register their physical activity level in detail with a questionnaire (physical activty registration) by using a QR code distributed at baseline. The participants will be reminded of this during follow-up by telephone or at the on-site training. To control subjective measures of adherance and physical activity level, 10% of the participants in the digital group will be randomly assigned to use an accelerometer for seven continous days during the intervention period.

## Participants, organization and collaborations

The project consortium will consist of members from SI (MD/Professor Arne Kolstad and PhD candidate Anna Solheim), HINN (Professor Stian Ellefsen, Dr. Anne Mette Rustaden, Dr. Håvard Hamarsland), Karolinska Institutet (Professor Anna Krook), «Aktiv mot kreft» (CEO Helle Aanesen), «Pusterommet Gjøvik» (Physiotherapist/ CEO Aina Strøm) and Aktivklinikken Lillehammer (Physiotherapist Lene Melgård Sæther). The project consortium consits of renowed researchers from the fields of medicine, physiology and health/exercise, and the group has broad experience with human RCTs, the entire research methods included in the project, as well as physical, psychological and metabolical/biological/physiological analyses. Professor Kolstad is senior consultant and specialist in oncology and professor of medicine at NTNU. He has broad experience as principal investigator and leader of multiple clinical trials, both early phase exploratory studies and international multi-center studies. Professor Ellefsen has broad experience with human RCTs, as well as with physiological and molecular biological analyses. He is initiative taker and leader of the Trainome biobank and has led the research projects in lifestyle therapy for more than a decade. Ass.Professor Rustaden is Physiotherapist and have experience with RCTs including resistance exercise and lifestyle therapy targeting different patients groups. Ass.Professor Hamarsland is the leader of the Trainome research group at the Faculty of Health- and Social Sciences. The PhD candidate Anna Solheim will attend the PhD program HELVEL, at HINN. Supervision will be organized as a team effort between Ass.Professor Rustaden (main), Professor Kolstad (co) and Professor Ellefsen (co). Ass.Professor Hamarsland and Professor Krook will be members of the project group and specifically assist with the physiological and biological assessments. To ensure the feasibility of the project, an Intervention Oversight Committee (IOC) will be established, consisting of key members from the consortium (Rustaden, Kolstad, Solheim, Aanesen, Strøm and Sæther) and user representatives (n=2, recruited through «Aktiv mot kreft» prior to study onset).

**Plan for activities, visibility and dissemination**

Figure 3 shows an overview over project milestones during the 3 years period. For detailed overview of the activities, see the application form. Under supervision of the project consortium, the PhD candidate will be responsible for leading, organizing and conducting the RCT, including organization of testing/sampling, preparing and organizing the exercise program according to protocol (in cooperation with «Aktiv mot kreft») and data collection. The interventions will be made possible by cooperation with physiotherapists at Gjøvik and Lillehammer, and master- and bachelorstudents at HINN will assisst during assessments and intervention. The PhD candidate will also need to cooperate with and assist Professor/MD Kolstad with recruitment and inclusion of patients.

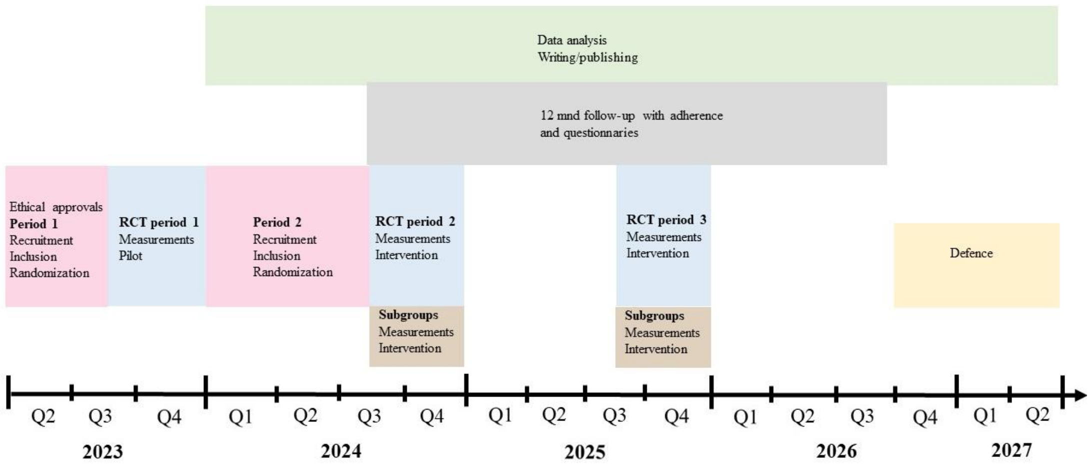


Figure 3: Timeline/milestones in the project. The main RCT includes patients after cancer treatment. Subgroups includes patients in active cancer treatment and participants living further than 20 km from the on-site training location.

The project and its results will be of broad interest to the national and international audience, both among researchers, physioterapists and other healthcare workers, the patient group and the public. It will bring out valuable knowledge on cancer rehabilitation that effectively can be implemented to the patients across the country, despite their domicile. In order for outcomes measures to be successfully implemented into clinical practice, the results must be disseminated in an appropriate manner. To achieve this, the IOC group will take a broad range, from publication in peer-reviewed open-access medical and exercise related journals, to presentations at medical and exercise science congresses and seminars. The IOC group will together with "Aktiv mot kreft" develop and outsource courses to physiotherapists and other healthrelated professionals, and further develop digital exercise solutions. In addition, IOC will launch an effective media and social media strategy, to reach out all over the country.

**Plan for implementation**

Results from the project will provide novel insight into the efficacy of digital resistance exercise on cancer patients for relieving a variety of functionally, mentally and metabolically health concerns. Further, the project will provide insight into exercise adherence with digital home-based exercise, which is a cost-effective and flexible way of distributing exercise to cancer patients all over the country, included patients living far away from clinics and other exercise locations. This insight will be valuable for both researchers, health-care providers, “Aktiv mot kreft” and other organizations distributing physical activity and the patient group alike. The projects long-term aim is to establish safe and standardized digital home-based exercise protocols for cancer patients after treatment, which immediately can be implemented in the clinical practice on a national level. To achieve this, an execute dissemination strategy targeting a broad audience will be developed, including academic and clinical communities, public and social media, politicians and organizations. First, results we will be published in open-access peer-reviewed scientific journals and presented at medical and sports related congresses/seminars (regionally, nationally and internationally). Secondly, for outcome measures to be successfully implemented into clinical practice and to bridge the gap between hospitals/oncologists and physiotherapists/exercise instructors in the public health service, we will in cooperation with “Aktiv mot kreft” develop and implement courses and workshop and initiate meeting points. In addition, together with “Aktiv mot kreft” we will develop and update the website (www.aktivmotkreft.no/pusterommetlive) and initiate seminars. Finally, to reach out to patients and therapists all over the country, we will develop and implement a media and social media strategy.

**User involvement**

The project group will assemble a resource group for user involvement, which together with the IOC will have regular meetings, and thus be involved in the planning of the project, implementation and evaluation. The resource group will consist of two cancer patients, CEO “Aktiv mot kreft”, two physiotherapists from “Pusterommet” Gjøvik, and two represents from the department of oncology at SI (one medical doctor and the head of the department).

**Ethical considerations**

The project will not start without approval from the Regional Committee for Medical Research Ethics, Norway. All participants will need to sign a written consent statement before entering the study. They will be informed that all participation is voluntary and that they at any time can drop out of the study, without justification. All procedures in the project will follow the World Medical Association Declaration of Helsinki, and the project protocol will be registrered at www.clinicaltrials.gov and www.helsenorge.no/kliniske-studier prior to initiation. All data will be stored and processed de-identified. First, during data collection, the data will be stored temporary in a project folder in the access regulated server EduCloud (with HINN Feide account). Secondly, the PhD student will transfer all data to Services for Sensitive Data (TSD) for storage. The blood samples will be coded and stored in the biobank

Trainome (REK-id:2013/2041) at HINN Lillehammer. The participants will get access to their own data on request.

Due to the number of positive health outcomes from resistance exercise, functionally, mentally and metabolically, the patiens utility value of participation in the project is considered good. This is regardless of group allocation, since both groups will be offered the same exercise program. However, to minimize the risk for healthrelated events during testing and/or exercise, the medical doctors involved in recruitment will strictly follow the exclusion criterias during study enrolement. In addition, all personnel included in the testing procedures and training, will be trained to handle unwanted incidents, and prior to the physical testing and exercise, the participants will be asked to fill out a self-declaration about their health, including questionnaire about symptoms during physical activity (e.g. chest pain, dyspnoea, dizziness). If the form reveal uncertainty regarding participation in the study, MD Arne Kolstad will be consulted and make the final decision. An automated external defibrillator is placed close to the locations where all testing and training will be performed. In addition, if unwanted symptoms arise during the exercise, in relation to any part of the project, they will be encouraged to contact the MD Arne Kolstad or their medical doctor for further follow-up and if necessary, withdrawal from the study. All assessments will follow standardized procedures, and experienced and trained personnel will complete all assessments. Anne Mette Rustaden (physiotherapist) will be present during all the physical assessments. During the venous blood sampling, the participants may experience discomfort. However, this invasive method is not associated with any risk. The DEXA scan is based on x-ray technology, but the radiation dose is negligible (Bazzocchi et al 2016). Participants may also experience discomfort and/or exhaustion during the intervention and physical assessments. Importantly, this is normal and appropriate experiences and purposes from exercise, and the instructors will constantly inform and instruct the participants in regard of intensity and exhaustion. Primarily, the present study is aimed to help cancer patients to improve their functional, mental and metabolic health after finishing active cancer treatment, and we are not looking for health challenges. However, should we discover something that deviates from expected findings and/or something that give us suspicion of health challenges, we will initiative further medical follow-up. The participants will then be contacted by the medically responsible in the study (Arne Kolstad) or by Physiotherapist Anne Mette Rustaden. Participants who wants to withdraw from the study are given the opportunity to voluntarily answer why they do not want to continue the study.

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